

9 Periodicity

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Across Period 3 nuclear charge increases and atomic radius generally decreases as shielding changes little; first ionisation energy broadly increases with subshell/pairing exceptions. Oxides change from basic metallic through amphoteric to acidic non-metallic. Explain anomalies with orbital energy and electron repulsion, not a blanket memorised trend.

Worked example: follow the reasoning

Al's first ionisation energy is lower than Mg's because the removed electron is in higher-energy 3p, not Mg's 3s.

Quick check · multiple choice

A Which oxide is generally more acidic?

[1]

A SO_3

B Na_2O

C MgO

D Al_2O_3 only

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain why first ionisation energy of S is lower than P despite greater nuclear charge.

[2]

Answers and explanation

Quick check: A. SO_3

Question 1 [2 marks]

In sulfur one 3p orbital has paired electrons [1]; inter-electron repulsion makes one easier to remove than the unpaired 3p electron in phosphorus [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.